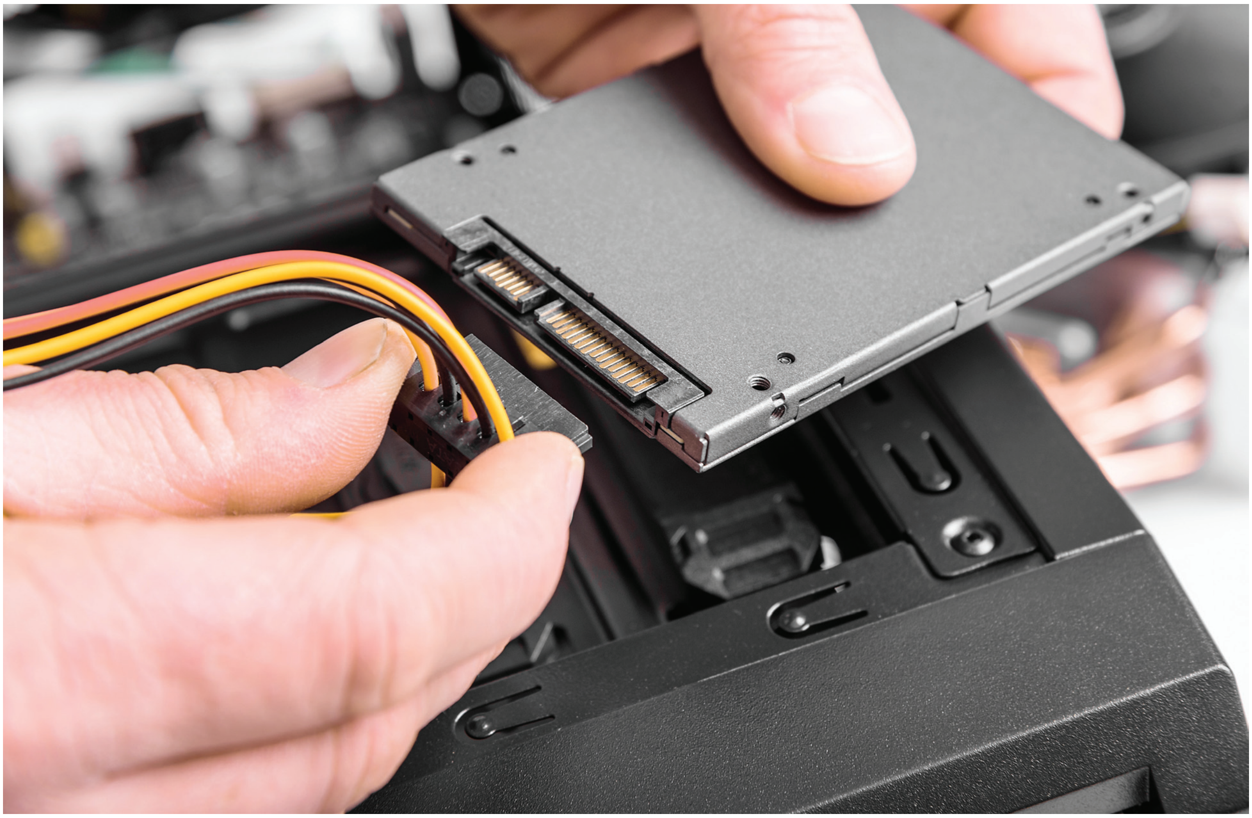
Unleash the Untapped Potential of Memory Intelligence



Over the past few decades, innovative memory architectures have shaped memory design for various physical devices. This change continues and is particularly interesting from the perspective of artificial intelligence, the Internet of Things, and robotics.

System development has come a long way over the years. The process began right when computers were being designed. In a system, the critical components are the processor and memory. When both of these are efficient, you get more processing power and better bandwidth for memory. Already a lot has happened on the processing side with the central processing units (CPUs) evolving into graphics processing units (GPUs) and later transitioning into neural processing units (NPUs).

This has come into effect in mobile phones with the introduction of NPUs for faster artificial intelligence (AI) processing. It seems the maximum limit has been reached there. Now the focus is shifting towards strengthening the memory efficiency. This is where 3D SRAMs and stack DSRAMs are coming in. “I will not be surprised if intelligent memory starts coming in the future. We have not reached that point yet, but we have come to a stage where we are thinking of pipelined design in software,” says Nikhil Bhaskaran, founder, Shunyaos.

Neural networks

Neural networks certainly hold promise in developing intelligent memory for faster data compute and efficiency. Huge datasets require frequent memory access. Doing that in today’s memory system creates high inefficiency, leading to speed loss. To overcome that, AI on edge offers a viable solution where a fast output can be obtained. “This correlates with testing the limits of a system, but that is when things start getting fixed,” says Nikhil.